

Goutham Vignesh

Senior AI Engineer | Multi-Agent AI Systems | AI Infrastructure

Applied AI Engineer with 6+ years of experience building and designing production-scale multi-agent AI systems. Specialized in AI infrastructure, inference optimization, multimodal AI, and cloud-native ML deployment. Experienced in designing and deploying agentic workflows, scalable inference systems, semantic search, and multimodal AI applications used in enterprise environments.

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WORK EXPERIENCE

01/2024 - Present	AI RESEARCH ENGINEER - APPLIED R&D UNIT Quantiphi Bangalore	- Architected and deployed a production-grade multi-agent AI platform using LangGraph, MCP, and Google ADK, enabling scalable tool orchestration, reusable agent skills, and enterprise workflow automation. - Led ML engineering for the IQVIA engagement by designing LangGraph-based workflows to automate clinical trial data review and accelerate delivery timelines. - Led the design and delivery of AI solutions across healthcare, document intelligence, and developer productivity domains, driving adoption of LLMs, agentic workflows, and multimodal AI systems. - Built AI coding assistants for Quantiphi's Codeaira platform with agent orchestration and skill-planning capabilities, reducing manual effort by 60%. - Architected and deployed IDPFlow, a production-grade multi-agent AI platform on Gemini Enterprise enabling multimodal document intelligence, tool orchestration, workflow composition, and human-in-the-loop execution, delivered as a scalable Agent-as-a-Service platform.
04/2023 - 01/2024	RESEARCH ENGINEER (GEN AI) SciSpace Bangalore	- Built cost-efficient LLM fine-tuning pipelines using PEFT and LoRA, reducing training costs by 75%. - Optimized open-source LLM deployment using vLLM, NVIDIA Triton, and quantization techniques to improve inference scalability and latency. - Applied model compression and quantization strategies for efficient LLM deployment on resource-constrained infrastructure.
12/2019 - 03/2023	MACHINE LEARNING ENGINEER Vakilsearch Chennai	- Built RAG-based document retrieval and semantic search systems using Pinecone and Weaviate for legal document intelligence. - Developed intelligent document processing pipelines combining OCR, OpenCV, and YOLO-based object detection for document classification and information extraction, achieving 85% accuracy in address-proof verification workflows. - Built scalable ML deployment pipelines and APIs for production model serving, improving deployment efficiency and system reliability by 30%.

PUBLICATIONS & RESEARCH

- **Stop Thinking, Start Looking:** Efficient Post-Training for Multimodal Document Question Answering via Reasoning-Free Alignment — **ICML 2026 EMMQA Workshop**
- **IDPFlow:** A No-Code Framework for Multimodal Intelligent Document Processing — **ACM Multimedia 2025**
- **DocAnnot:** Accelerating Key Information Extraction Dataset Creation — **Book Chapter, 2026**
- **DocGenie:** A Framework for High-Fidelity Synthetic Document Generation — **CVPR Workshop 2025**
- **SciSpace Copilot:** Intelligent Reading Assistance for Researchers — **AAAI 2024**
- **SciSpace Literature Review:** AI for Scientific Discovery — **ECIR 2024**

SKILLS

Programming & Frameworks

- Python, FastAPI, REST APIs, Git, Linux

AI & Machine Learning

- PyTorch, TensorFlow, Transformers, Reinforcement Learning, Ensemble Learning, Computer Vision

LLM & Agentic Systems

- LLMs, RAG, LangGraph, MCP, Google ADK, Multi-Agent Systems, PEFT, LoRA

Multimodal & Document AI

- OCR, Document Intelligence, OpenCV, YOLO, Semantic Search, Knowledge Graphs

Deployment & Infrastructure

- AWS, GCP, Azure, Docker, Kubernetes, NVIDIA Triton, vLLM

Databases & Vector Stores

- MongoDB, Pinecone, Weaviate

EDUCATION

2020	Bachelor of Engineering
Sri Krishna College of Technology, affiliated to Anna University Coimbatore, India	Computer Science Engineering

LANGUAGES

- **English - Professional Working Proficiency**

WEBSITES

[LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Portfolio](#) | [ORCID](#)